$$\frac{3\times\pi}{8\times 12}\left[\left[0+\sin\frac{\pi}{12}\right]+2\left[\sin\frac{\pi}{4}\right]+3\times\left[\sin\frac{\pi}{12}+\sin\frac{\pi}{6}+\sin\frac{\pi}{3}+\sin 5\frac{\pi}{12}\right]\right]=$$

Ans.

Q. Given that

x	4.0	4.2	4.4	4.6	4.8	5.0	5.2
$\log x$	1.3863	1.4351	1.4816	1.5261	1.5686	1.6094	1.6487

Evaluate $\int_4^{5.2} \log x\,dx$ by (a) Trapezoidal rule (b) Simpson's (1/3) rule

(c) Simpson's (3/8) rule.

Gaussian Quadrature

$\int_a^b f(x)$ 3 step process.

Step 1: change the interval $[a,b]$ into $[-1,1]$

using the linear eqn: $x = \frac{b-a}{2}t + \frac{b+a}{2} \Rightarrow dx = \frac{b-a}{2}dt$

Step 2: Substitute x and dx in the original integral.

i.e $\int_a^b f(x)\,dx = \int_{-1}^{1} F(t)\,dt$

Step 3: Evaluating the integral.

One point form ($n=0$): $\int_{-1}^{1} F(t)\,dt = 2F(0)$

Two point form ($n=1$): $\int_{-1}^{1} F(t)\,dt = F\left(-\frac{1}{\sqrt{3}}\right) + F\left(\frac{1}{\sqrt{3}}\right)$

Three point form ($n=2$): $\int_{-1}^{1} F(t)\,dt = \frac{1}{9}\left[5F\left(-\sqrt{\frac{3}{5}}\right) + 8F(0) + 5F\left(\sqrt{3/5}\right)\right]$

Q. Evaluate integral $I = \int_0^1 \frac{dx}{1+x}$ using Gauss 3 point formula.

→ Step 1: change interval $[a,b] = [0,1]$ into $[-1,1]$

$$x = \frac{b-a}{2}t + \frac{b+a}{2}$$

$a = 0$
$b = 1$

$$x = \frac{1-0}{2}t + \frac{1+0}{2} = 0.5t + 0.5 \quad \text{or} \quad \frac{1}{2}(t+1)$$

$x \Rightarrow \frac{1}{2}t + \frac{1}{2} \Rightarrow \frac{t+1}{2}$

$\Rightarrow dx = \frac{dt}{2}$

Step 2: $I = \int_0^1 \frac{dx}{1+x} = \int_{-1}^1 \frac{dt/2}{1+\frac{t+1}{2}} = \int_{-1}^1 \frac{dt}{t+3} = [\log t+3]_{-1}^1$

$= \log 1+3 = \log -1+3 = 3 = \log 1 - \log(-1) = \log \frac{-1}{1}$

Gauss- three point form,

$$\int_{-1}^1 F(t)\,dt = \frac{1}{9}\left[5F\left(-\sqrt{3/5}\right) + 8F(0) + 5F\left(\sqrt{3/5}\right)\right]$$

$\Rightarrow F(t) = \frac{1}{t+3}$

$$\Rightarrow \int_{-1}^1 \frac{1}{t+3}\,dt = \frac{1}{9}\left[5 \times \frac{1}{-\sqrt{3/5}+3} + 8 \times \frac{1}{0+3} + 5 \times \frac{1}{\sqrt{3/5}+3}\right]$$

$$= \frac{1}{9}\left[2.246 + 2.666 + 1.324\right] = \underline{0.6931}$$

Q. Evaluate $I = \int_1^2 \frac{2x}{1+x^4}\,dx$ using 1 point, 2 point & 3 point form

→ $x \in [a,b] = [1,2]$

$$x = \frac{b-a}{2}t + \frac{b+a}{2} = \frac{2-1}{2}t + \frac{2+1}{2} \Rightarrow \frac{1}{2}t + \frac{3}{2} = \frac{t+3}{2}$$

$x = \frac{t+3}{2} \Rightarrow dx = \frac{dt}{2}$

Step 2: $I = \int_{1}^{2} \frac{2x}{1+x^4}\,dx = \int_{-1}^{1} \frac{2x\left(\frac{t+3}{2}\right)}{1+\left(\frac{t+3}{2}\right)^4}\,(dt/2)$

$= \int_{-1}^{1} \frac{t+3}{\frac{16+(t+3)^4}{16}} \cdot \frac{dt}{2} = \int_{-1}^{1} \frac{t+3}{\frac{16+(t+3)^4}{8}} \cdot \frac{dt}{2} = \int_{-1}^{1} \frac{8(t+3)}{16+(t+3)^4}\,dt$

$\therefore F(t) = \frac{8(t+3)}{16+(t+3)^4}$

2point
0.5434

One point form: $\int_{-1}^{1} F(t)\,dt = 2F(0)$

3point
0.5406

$\Rightarrow \int_{-1}^{1} \frac{8(t+3)}{16+(t+3)^4}\,dt = 2 \times \frac{8(0+3)}{16+(0+3)^4} = 0.4948$

Two point form: $\int_{-1}^{1} F(t)\,dt = F\left(-\frac{1}{\sqrt{3}}\right) + F\left(\frac{1}{\sqrt{3}}\right)$

$\Rightarrow \int_{-1}^{1} \frac{8(t+3)}{16+(t+3)^4}\,dt = \frac{8\left(-\frac{1}{\sqrt{3}}+3\right)}{16+\left(-\frac{1}{\sqrt{3}}+3\right)^4} + \frac{8\left(\frac{1}{\sqrt{3}}+3\right)}{16+\left(\frac{1}{\sqrt{3}}+3\right)^4} = 0.5433$

Three point form: $\int_{-1}^{1} F(t)\,dt = \frac{1}{9}\left[5F\left(-\sqrt{3/5}\right) + 8F(0) + 5F\left(\sqrt{3/5}\right)\right]$

$\Rightarrow$

Ordinary Differential Equations

(1) Euler's Method

$\frac{dy}{dx} = F(x, y)$ with some initial condition

$y(x_0) = y_0$; To find $y(x_n)$

using the iteration $\boxed{y_{n+1} = y_n + h f(x_n, y_n)}$

Q. Solve the initial value problem $y' + 0.2y = 0$; $y(0) = 5$, $h = 0.2$ using Eulers method taking 5 steps.

→ $x_0 = 0$ $\quad y_0 = 5$ $\quad y' + 0.2y \Rightarrow y' = -0.2y = F(x, y)$

$x_1 = x_0 + h = 0 + 0.2 = 0.2$ $\quad y_1 = y_0 + h f(x_0, y_0) = 5 + 0.2 \times (-0.2 \times 5) = 4.8$

$x_2 = 0.4$ $\quad y_2 = 4.8 + 0.2 \times (-0.2 \times 4.8) = 4.608$

$x_3 = 0.6$ $\quad y_3 = 4.608 + 0.2 \times (-0.2 \times 4.608) = 4.42368$

$x_4 = 0.8$ $\quad y_4 = 4.42368 + 0.2 \times (-0.2 \times 4.42368) = 4.2467328$

$x_5 = 1$ $\quad y_5 = 4.2467328 + 0.2 \times (-0.2 \times 4.2467328) = 4.07686$

≈ 4.07

Q. Use Euler's method ; solve with $h = 0.025$, compute the value of $y(0.1)$ from $y' = x - y^2$; $y(0) = 1$

→ $x_0 = 0$

$y_0 = 1$

Initial condition, $x_0 = 0$, $y_0 = 1$; $h = 0.025$, $F(x, y) = x - y^2$

$x_0 = 0$ $\quad y_0 = 1$

$x_1 = 0.025$ $\quad y_1 = y_0 + h \times F(x_0, y_0) = 1 + 0.025 \times (0 - 1^2) = 0.975$

$x_2 = 0.05$ $\quad y_2 = 0.975 + 0.025 \times (0.025 - 0.975^2) = 0.95186$

$x_3 = 0.075$ $\quad y_3 = 0.951 + 0.025 \times (0.05 - 0.951^2) = 0.93046$

$x_4 = 0.1$ $\quad y_4 = 0.929 + 0.025 \times (0.075 - 0.929^2) = 0.91069$

$x_5 = 0.125$

$x_6 = 0.15$

How.

Q. Using Euler's method, solve $y' = x + y$; $y(0) = 2$, for $x = 0.2$ (ans: 2.43)

[here h is chosen according to marks ⇒ 1 mark → 1 iteration] for 2M ⇒ $h = 0.1$, for 5M = 0.05

(2)

Runge-Kutta Method [R-K Method]

Second order R-K method

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = hf(x_n, y_n)$

$k_2 = hf(x_n + h, y_n + k_1)$

Q. Applyig R-K method of 2nd order, solve initial value problem $\frac{dy}{dx} = x + y$ $y(0) = 1$. Find y at $x = 0.2$, taking $h = 0.1$

→ $x_0 = 0$, $y_0 = 1$, $h = 0.1$, $\frac{dy}{dx} = x + y = F(x, y)$

$x_0 = 0$ $\quad y_0 = 1$

$x_1 = 0 + 0.1 = 0.1$ $\quad y_1 = y_0 + \frac{1}{2}(k_1 + k_2)$

$x_2 = 0.2$

$k_1 = hf(x_0, y_0) = 0.1 \times (0 + 1) = 0.1$

$k_2 = hf(x_0 + h, y_0 + k_1) = 0.1 \times ((0 + 0.1) + (1 + 0.1))$

$= 0.12$

$y_1 = 1 + \frac{1}{2}(0.1 + 0.12) = 1.11$

$y_2 = y_1 + \frac{1}{2}(k_1 + k_2)$, $\quad k_1 = 0.1 \times (0.1 + 1.11) = 0.121$

$k_2 = 0.1 \times ((0.1 + 0.1) + (1.11 + 0.121)) = 0.1431$

$y_2 = 1.11 + \frac{1}{2}(0.121 + 0.1431) = 1.24205$

Fourth Order R-K method

$y' = f(x, y)$; $y(x_0) = y_0$, h is given

$$y_{n+1} = y_n + \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$$

where, $k_1 = hf(x_n, y_n)$

$k_2 = hf(x_n + h/2, y_n + k_1/2)$

$k_3 = hf(x_n + h/2, y_n + k_2/2)$

$k_4 = hf(x_n + h, y_n + k_3)$

Q. Solve the differential eqn by using 4th order R-K method for $y(0.6)$ from $\frac{dy}{dx} = x^2+y$; $y(0)=3$ & taking $h=0.3$

→ $\frac{dy}{dx} = x^2+y = f(x,y)$ $\quad y(0)=3 \Rightarrow x_0 = 0$ & $y_0 = 3$, $h=0.3$

1st iteration

$x_1 = x_0+h = 0+0.3 = 0.3$

Find $y(0.3)$ on y_1

$y_{n+1} = y_n + \frac{1}{6}[k_1+2k_2+2k_3+k_4]$

$y_1 = y_0 + \frac{1}{6}[k_1+2k_2+2k_3+k_4]$

$k_1 = hf(x_0,y_0) = 0.3 \times (0^2+3) = 0.9$

~~$k_2 = 0.3 \times ($~~

$k_2 = 0.3 \times f\left(x_0+\frac{0.3}{2}, y_0+\frac{k_1}{2}\right)$

~~$x_1 = 0.3$~~ $\quad y_1 =$

~~$x_2 = 0.6$~~

~~$x_3 = 0.9$~~

~~$x_4 = 1.2$~~

$\Rightarrow k_2 = 0.3 \times (0.15^2+3.45)$ $= 0.3 \times f(0.15, 3.45)$

$= 1.041$

$k_3 = 0.3 \times f\left(0+\frac{0.3}{2}, 3+\frac{1.041}{2}\right) = 0.3 \times f(0.15, 3.52) = 0.3 \times (0.15^2+3.52)$

$= 1.062$

$k_4 = 0.3 \times f(0+0.3, 3+1.062) \Rightarrow 0.3 \times (0.3^2+4.062) =$ ~~0.9456~~ 1.2456

$\therefore y_1 = 3 + \frac{1}{6} \times [0.9 + (2 \times 1.041) + (2 \times 1.062) +$ ~~0.9456~~ $1.2456]$

$=$ ~~4.0086~~ 4.058

2nd iteration

$y_2 = y_1 + \frac{1}{6}[k_1+2k_2+2k_3+k_4]$

$x_1 = 0.3$

$y_1 = 4.0586$

$k_1 = 0.3 \times (0.3^2+4.0586) = 1.2445$

$k_2 = 0.3 \times \left(\left(0.3+\frac{0.3}{2}\right)^2 + 4.0586 + \frac{1.2445}{2}\right) = 1.465$

$k_3 = 0.3 \times \left(\left[0.3+\frac{0.3}{2}\right]^2 + \left[4.0586+\frac{1.465}{2}\right]\right) = 1.498$

$k_4 = 0.3 \times \left[(0.3+0.3)^2 + (4.0586+1.498)\right] = 1.774$

$y_2 = 4.0586 + \frac{1}{6}[1.2445 + (2 \times 1.465) + (2 \times 1.498) + 1.774] = 5.549$

Q. Find y at $x=0.1$ using 4th order R-K method, given that $y' - yx^2 = 0$; $y(0) = 1$

→ given, $y' - yx^2 = 0 \Rightarrow y' = yx^2 \Rightarrow f(x,y)$

$x_0 = 0$ $\quad y_0 = 1$

$x_1 = 0.1$ $\quad y(0.1) = ?$

choose $h = 0.1$

1st iteration

$$y_1 = y_0 + \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$$

$k_1 = 0.1 \times (1 \times 0^2) = 0$

$k_2 = 0.1 \times \left[\left(0 + \frac{0.1}{2}\right) \times \left(1 + \frac{0}{2}\right)\right] = 0.005$

$k_3 = 0.1 \times \left[\left(0 + \frac{0.1}{2}\right) \times \left(1 + \frac{0.005}{2}\right)\right] = 0.00501$

$k_4 = 0.1 \times [(0+0.1) \times (1 + 0.00501)] = 0.01005$

$y_0 \Rightarrow y_1 = 1 + \frac{1}{6}[0 + (2 \times 0.005) + (2 \times 0.00501) + 0.01005]$

$= 1.005$

(3) Taylor's Series Method

$$\frac{dy}{dx} = F(x,y) \quad \text{——— ①}$$

$y(x_0) = y_0$

If $y(x)$ is the exact solution of eqn ①, ~~subst~~ subject to given condition then the Taylor series of $y(x)$ around $x = x_0$ is given by

② ——— $y_1 = y(x) = y_0 + (x - x_0)\, y_0' + \frac{(x-x_0)^2}{2!} y_0'' + \frac{(x-x_0)^3}{3!} y_0''' + \cdots$

If $x - x_0 = h$ in eqn ②,

$$y_1 = y(x) = y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{IV} + \cdots$$

Similarly, Taylor's series expansion for $y(x)$ around $x = x_1$

$$y_2 = y_1 + h y_1' + \frac{h^2}{2!} y_1'' + \frac{h^3}{3} y''' + \frac{h^4}{4!} y^{IV} + \cdots$$

In general,

$$y_{n+1} = y_n + hy_n' + \frac{h^2}{2!} y_n'' + \frac{h^3}{3!} y_n''' + \dots\dots$$

Q. Solve $\frac{dy}{dx} = x+y$ by Taylor's series method, start from $x=1$, $y=0$ and carry to $x=1.2$ ~~wh~~ with $h=0.1$

→ Initial condition ⇒ $x=1$, $y=0$ $h=0.1$

$x_0 = 1$ $y_0 = 0$

$x_1 = 1.1$ $y_1(x_1) = ?$

$x_2 = 1.2$ $y_2(x_2) = ?$

given,

$\frac{dy}{dx} = x+y \Rightarrow y' = x+y$ $y_0' = 1+0 = 1$

$y'' = 1+y'$ $y_0'' = 1+1 = 2$

$y''' = y''$ $y_0''' = 2$

$y^{iv} = y'''$ $y_0^{iv} = 2$

$$y_1 = y(1.1) = y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{iv} + \dots\dots$$

$$= 0 + (0.1 \times 1) + \left(\frac{0.1^2}{2!} \times 2\right) + \left(\frac{0.1^3}{3!} \times 2\right) + \left(\frac{0.1^4}{4!} \times 2\right)$$

$$= 0.1103$$

~~$y_2 = y(1.2) = y_1 + hy_1'$~~ Now, $x_1 = 1.1$, $y_1 = 0.1103$

$y_1' = 1.1 + 0.1103 = 1.2103$

$y_2'' = 1 + 1.2103 = 2.2103$

$y_2''' = 2.2103$

$y_2^{iv} = 2.2103$

$$y_2 = y(1.2) = y_1 + h y_1' + \frac{h^2}{2!} y_1'' + \frac{h^3}{3!} y_1''' + \frac{h^4}{4!} y_1^{IV} +$$

$$= 0.1103 + (0.1 \times 1.2103) + \left(\frac{0.1^2}{2!} \times 2.2103\right) + \left(\frac{0.1^3}{3!} \times 2.2103\right) + \left(\frac{0.1^4}{4!} \times 2.2103\right)$$

$$= 0.2428$$

Q. Find Taylor's series solution of the eqn $\frac{dy}{dx} = x - y^2$; $y(0) = 1$.
Find y at $x = 0.1$ and $x = 0.2$

→ $x_0 = 0$, $y_0 = 1$, $h = 0.1$

$(u \cdot v)' = u \cdot v' + v \cdot u'$

$x_1 = x_0 + h = 0.1 \Rightarrow y(0.1) = ?$

$x_2 = 0.2 \Rightarrow y(0.2) = ?$

$\frac{dy}{dx} = y' = x - y^2$ $\quad y_0' = 0 - 1^2 = -1$

$y'' = 1 - 2y \cdot y'$ $\quad y_0'' = 1 - 2 \times 1 \times -1 = 3$

$y''' = -2[y \cdot y'' + y' \cdot y']$ $\quad y_0''' = -2[1 \times 3 + -1 \times -1] = -8$

$y^{IV} =$

Boundary Value Problems

A boundary value problem comprises a differential eqn subject to a given set of boundary conditions. The no. of boundary conditions is equal to order of ordinary diff. eqn.

Finite Difference Method

It is used to solve a BVP. FDM reduces the ODE into a system of linear equations :

→ Every derivative occuring in the equation as well as in boundary conditions

are replaced by finite difference approximation

→ The eqn is then reduced to a set of linear system of eqns which are then solved by using standard methods.

The Finite difference approximations are,

for $y'(x) = \frac{y(x+h) - y(x-h)}{2h}$

If $x = x_i$ $\quad y_i' = \frac{y_{i+1} - y_{i-1}}{2h}$

lly $\quad y''(x) = \frac{y(x+h) - 2y(x) + y(x-h)}{h^2}$

i.e $\quad y_i'' = \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$

Q. Using FDM, solve for y given the differential eqn $y'' + y - 1 = 0$ where $x \in (0,1)$ and boundary condition $y(0) = y(1) = 0$. Taking $h = \frac{1}{4}$.

→ Given, $y'' + y - 1 = 0$

BCS Bounday condition, $y(0) = 0$ & $y(1) = 0$

final values, $x_n = 1$, $y_n = 0$

Step 1

$x_0 = 0$ $\quad y_0 = 0$

$x_1 = 0 + \frac{1}{4} = 0.25$ $\quad y_1 = ?$

$x_2 = 0.5$ $\quad y_2 = ?$

$x_3 = 0.75$ $\quad y_3 = ?$

$x_4 = 1$ $\quad y_4 = 0$

Step 2 ODE in FDM at $x = x_i$

$y_i'' + y_i - 1 = 0 \Rightarrow$ $\frac{y_{i+1} - y_{i-1}}{2}$

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + y_i - 1 = 0$$

$$\Rightarrow y_{i+1} - 2y_i + y_{i-1} + y_i h^2 - h^2 = 0$$

$$\Rightarrow y_{i+1} + y_i (h^2 - 2) + y_{i-1} - h^2 = 0 \quad \text{——— (I)}$$

To find y_1 :

$\because y_1 = y_i \Rightarrow$ i.e $i = 1$

$$\Rightarrow y_2 + (h^2 - 2) y_1 + y_0 - h^2 = 0 \qquad [h = 0.25, \; y_0 = 0]$$

$$\Rightarrow y_2 + (0.25^2 - 2) \times y_1 + 0 - 0.25^2 = 0$$

$$y_2 - 1.9375 y_1 - 0.0625 = 0 \quad \text{——— (1)}$$

To find y_2, put $i = 2$ in (I)

$$y_3 + (h^2 - 2) y_2 + y_1 - h^2 = 0 \Rightarrow y_3 - 1.9375 y_2 + y_1 - 0.0625 = 0 \quad \text{——— (2)}$$

To find y_3, put $i = 3$ in (I)

$$y_4 + (h^2 - 2) y_3 + y_2 - h^2 = 0 \Rightarrow h = 0.25, \; y_4 = 0$$

$$-1.9375 y_3 + y_2 - 0.0625 = 0 \quad \text{——— (3)}$$

solving (1), (2) & (3), $y_3 = -0.1047$, $y_2 = -0.1403$, $y_1 = -0.1047$

Q. Solve $\frac{d^2y}{dx^2} - y = 0$; B.C.s $y(0) = 0$, $y(3) = 0$ and $h = 1$

$x_0 = 0$	$x_1 = 1$	$x_2 = 2$	$x_3 = 3$
$y_0 = 0$	$y_1 = ?$	$y_2 = ?$	$y_3 = 0$

given eqn, $\frac{d^2y}{dx^2} - y = 0 \Rightarrow y'' - y = 0$

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{} - y_i = 0 \quad \text{——— (I)}$$

$y_1 \leftarrow x \leftarrow y_2 \leftarrow 2y_1$

$$y_{i+1} - 2y_i + y_{i-1} - y_i h^2 = 0 \Rightarrow y_{i+1} - y_i (+2+h^2) + y_{i-1} = 0 \quad \text{—— (I)}$$

for y_1

(I) $\Rightarrow y_2 - y_1 (+2+1^2) + y_0 = 0$ $\quad h = 1$, $y_0 = 0$

$\Rightarrow y_2 + 3y_1 + 0 = 0 \Rightarrow y_2 - 3y_1 = 0$ —— (1)

for y_2

$y_3 - y_2 (+2+1^2) + y_1 = 0 \Rightarrow y_3 - 3y_2 + y_1 = 0$ —— (2) $\quad [y_3 = 3]$

$\Rightarrow -3y_2 + y_1 = 0$ —— (2)

solving (1) and (2),

$\Rightarrow 3 - 3y_2 + y_1 = 0$ —— (2)

$y_1 = 0$ $\quad y_2 = 0$

$y_1 = 0.375$, $y_2 = 1.125$

HW

Q.1) solve $y'' - y = x$ given that $y(0) = y(1) = 0$ using FDM dividing the interval into 4 equal parts.

[ans: $y_1 = -0.0349$, $y_2 = -0.0564$, $y_3 = -0.0501$]

Q 2) solve $y'' + xy' + xy = 0$, $h = 0.025$, $y(0) = 1$ & $y(1) = 4$

[ans: $y_1 = 1.8076$, $y_2 = 2.637$, $y_3 = 3.4063$]

Statistical Concepts & Tests

Random Variable (x)

A variable whose value depends on outcome of an experiment.

Two types:

- Discrete RV :- The values are countable.
  eg: No. of heads in 3 coin tosses - (0, 1, 2 or 3)
- Continuous RV :- values in an interval. Eg: height, weight etc.

Expectation, Mean, Variance

For discrete RV ; $x = x_1, x_2, \ldots, x_n$

$P(x) = P_1, P_2, \ldots, P_n$

Expectation of x | Mean x

$$\mu = E(x) = \sum_{i=1}^{n} x_i p_i$$

Variance of x

$$\sigma^2 = V(x) = E(x^2) - [E(x)]^2 \quad \text{where} \quad E(x^2) = \sum_{i=1}^{n} x_i^2 p_i$$

Standard deviation

$$\sigma = \sqrt{Var(x)}$$

properties

→ $E(x+y) = E(x) + E(y)$

→ $E(ax+b) = aE(x) + b$

→ $V(ax+b) = a^2 V(x)$

Q. A random variable 'x' has the following probability distribution

X :	-2	-1	0	1	2	3
P(x) :	1/10	k	1/5	2k	3/10	3k

Compute (a) k (b) Mean & variance of x (c) E (4x+5)
(d) V (4x+5)

→ (a) Sum of probability , $\Sigma P = 1$

$$\frac{1}{10} + k + \frac{1}{5} + 2k + \frac{3}{10} + 3k = 1 \quad \Rightarrow \frac{3}{5} + 6k = 1$$

$$k = \frac{1}{15}$$

(b) Mean of X , $\mu = E(x) = \Sigma x_i p_i = -2 \times \frac{1}{10} + -1 \times \frac{1}{15}$

$$= -2 \times \frac{1}{10} + -1 \times \frac{1}{15} + 0 \times \frac{1}{5} + 1 \times \frac{2}{15} + 2 \times \frac{3}{10} + 3 \times \frac{3}{15}$$

$$= \frac{16}{15} = \underline{1.067}$$

Var (x) , $\sigma^2 = E(x^2) - [E(x)^2]^2$

$$E(x^2) = \Sigma x_i^2 p_i = (-2)^2 \times \frac{1}{10} + (-1)^2 \times \frac{1}{15} + 0 + 1^2 \times \frac{2}{15} + 2^2 \times \frac{3}{10} + 3^2 \times \frac{3}{15}$$

$$= \frac{18}{5} = \underline{3.6}$$

$$\sigma^2 = E(x^2) - [E(x)]^2 = 3.6 - (1.067)^2 = \underline{2.46}$$

(c) E (4x+5)

$$E(ax+b) = a\,E(x) + b$$

$$E(4x+5) = 4\,E(x) + 5 = 4 \times 1.067 + 5 = \underline{9.268}$$

(d) V (4x+5)

$$V(aX+b) = a^2 V(X)$$

$$V(4X+5) = 4^2\, V(X) = 4^2 \times 2.46 = \underline{39.36}$$

Q. A RV X has the following probability function.

X :	0	1	2	3	4	5	6	7
P(X) :	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

(i) Find the value of k

(ii) Evaluate $P(X<6)$, $P(X \geq 6)$

(iii) $P(0<X<5)$

note: $P(x) \geq 0$

$\Sigma P(X) = 1$

→ (i) Since $\Sigma P(X) = 1$, $0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$10k^2 + 9k - 1 = 0$, $k = \underline{0.1}$

(ii) $P(X<6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$

$= \frac{81}{100} = \underline{0.81}$

$P(X \geq 6) = 1 - P(X<6) = \frac{19}{100} = \underline{0.19}$

(iii) $P(0<X<5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$

$= k + 2k + 2k + 3k = 0.1 + 2\times0.1 + 2\times0.1 + 3\times0.1$

$= \underline{0.8}$

Q. A die is tossed thrice. A success is getting 1 on 6 on a toss. Find the mean & variance of the no. of successes.

→ P[getting 1 on 6] = P[success] = $\frac{2}{6} = \frac{1}{3} = p$

failure, $q = \frac{4}{6} = \frac{2}{3}$

∴ ~~P[no success]~~ Probability of getting no success = $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$

Probability of 1 success & 2 failures $= {}^3C_1 \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{4C_1}{9}$

probability of 2 success & 1 failure $= {}^3C_2 \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2C_2}{9}$

probability of 3 successes $= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$

Mean $= \mu = \sum x_i p_i = 0 + 1 \times \frac{4}{9} + 2 \times \frac{2}{9} + 3 \times \frac{1}{27} = 1$

Variance $= \sigma^2 = \sum x_i^2 p_i - \mu^2 = \frac{2}{3} = 0.667$

$E(x^2) = \sum x_i^2 p_i = 0 + 1^2 \left(\frac{4}{9}\right) + 2^2 \left(\frac{2}{9}\right) + 3^2 \left(\frac{1}{27}\right) = \frac{5}{3} = 1.667$

Binomial Distribution

A binomial distribution model ~~or~~ counts the no. of successes over a fixed no. of independent trials. If 'n' is the total no. of independent trials such that for each trial 'p' is the probability of success and 'q' that of failure, 'x' is the no. of successes achieved in those 'n' trials. The probability Mass function (PMF),

$$P(x = x) = {}^nC_x \, p^x \, q^{n-x}$$

where, $x = 0, 1, 2, 3, \cdots, n$.

p^x : prob. of getting exactly 'x' successes.

$p + q = 1$

↳ If a variable x follows binomial distribution w/ parameters n & p, it is denoted by $X \sim b(n, p)$.

Since, sum of all possible prob. in a distribution = 1, we

i.e $\sum_{x=0}^{n} {}^nC_x \, p^x \, q^{n-x} = 1$, we can connect it to Binomial theorem,

$$(p+q)^n = \sum_{x=0}^{n} {}^nC_x \, p^x \, q^{n-x} = 1$$

imp ★

Derive mean and variance of Binomial Distribution

If $x \sim b(n,p)$ with pmf, $f(x) = {}^nC_x p^x q^{n-x}$ where $x = 0, 1, 2 \ldots n$ and $p+q=1$

$$\text{Mean}(\mu) = E(X) = \sum x_i p_i = \sum_{x=0}^{n} x \cdot {}^nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} x \cdot \frac{n!}{x!\,(n-x)!} \cdot p^x \cdot q^{n-x} \qquad (x! = x(x-1)!)$$

$$= \sum_{x=1}^{n} \frac{n!}{(x-1)!\,(n-x)!} \cdot p^x \cdot q^{n-x}$$

$$= \sum_{x=1}^{n} np \frac{(n-1)!}{(x-1)!\;n-1-(x-1)!} p^{x-1} q^{n-1-(x-1)}$$

put $x-1 = y$,

$$= np \sum_{y=0}^{n} \frac{(n-1)!}{y!\,(n-1-y)!} \cdot p^y q^{n-1-y}$$

$$= np \sum_{y=0}^{n} {}^{n-1}C_y p^y q$$

$$= np\,(p+q)^{n-1} = \underline{np}$$

$$\boxed{\mu = np}$$

new Q. Derive the variance $(\sigma^2 = npq)$

Q. The probability that a pen manufactured by a company will be defective is $1/10$. If 12 such pens are manufactured, find the probability that (a) exactly 2 will be defective.
(b) atleast 2 will be defective.
(c) none will be defective.

→ Prob. of defective pen $= 1/10 = 0.1$ $\quad n = 12, \; p = 0.1, \; q = 0.9$

prob of non-defective pen $= 1 - 0.1 = 0.9$

(a) prob. of exactly 2 will be defective $= {}^{n}C_x \, p^x \, q^{n-x}$

$= 12 \, C_2 \, (0.1)^2 (0.9)^{10} \quad (x = 2)$

$= \underline{\underline{0.2301}}$

(b) P [at least 2 will be defective] $= 1 - P$ [either none or one will be none defective]

i.e $= 1 - [12 C_0 \, (0.9)^{12} + 12 C_1 \, (0.1)(0.9)^{11}]$

case (i) zero defective
(ii) one defective

$= \underline{\underline{0.3409}}$

(c) P [none will be defective] $= 12 \, C_{12} \, (0.9)^{12}$ or $12 \, C_0 (0.1)^0 (0.9)^{12}$

$= \underline{\underline{0.2824}}$

Q. In 256 sets of 12 tosses of a coin in how many cases one can expect 8 heads and 4 tails.

→ x = no. of success out of total trials.

$P(\text{head}) = \frac{1}{2} \qquad P(\text{tail}) = \frac{1}{2}$

Prob. 8 heads & 4 tails in 12 trials, $P(X = 8) = 12 \, C_8 \, (1/2)^8 \cdot (1/2)^4$

$= \frac{495}{4096} = \underline{\underline{0.1208}}$

The expected no. of such cases in 256 sets $= 256 \times 0.1208 = \underline{\underline{30.93}} \approx \underline{\underline{31}}$

Q. The following data are the no. of seeds germinating out of 10 on a damp filter paper for 80 sets of seeds. Fit a binomial distribution.

x :	0	1	2	3	4	5	6	7	8	9	10
f :	6	20	28	12	8	6	0	0	0	0	0

→ Here, $n = 10$, $N = \Sigma f_i = 80$

$\therefore$ mean $= \dfrac{\Sigma x_i f_i}{\Sigma f_i} =$ ~~2.175~~ 2.175

$\mu = np$
$\sigma^2 = npq$

In Binomial distribution, mean $= np = 2.175$

$\Rightarrow p = \dfrac{2.175}{n} = \dfrac{2.175}{10} = 0.2175$

$q = 1 - p = 1 - 0.2175 = 0.7825$

$\therefore$ The binomial distribution to be fitted;

$N(p+q)^n = N \cdot \sum_{x=0}^{n} {}^nC_x p^x q^{n-x} = N\left[{}^nC_0 p^0 q^n + {}^nC_1 p q^{n-1} + \cdots\right]$

$= 80\left[{}^{10}C_0 (0.2175)^0 (0.7825)^{10} + {}^{10}C_1 (0.2175)^1 (0.7825)^9 + {}^{10}C_2 (0.2175)^2 (0.7825)^8 + \cdots {}^{10}C_{10} (0.2175)^{10} (0.7825)^0\right]$

$= 0.086 + 6.885 + 19.138 + 23.938 + 17.743 + 8.630 + 2.878 + 0.66 + 0.105$

→

x :	0	1	2	3	4	5	6	7	8	9	10
f :	6.9	19.1	24.0	19.8	8.6	2.9	0.7	0.1	0	0	0

Poisson Distribution

It is a distribution related to the probabilities of events which are extremely rare but which have a large no. of independent opportunities for occurance. Eg: The no. of persons born blind per year in a large city, • The no. of death by horse kick in an army corps. This distribution can be derived as a limiting case of binomial distribution by making 'n' very large and 'p' very small. ($n > 30$, p-small).

$$\text{Pmf } (x, \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \ldots$$

where λ is the parameter

Derivation of Mean and Variance of poisson distribution

$x! = x(x-1)!$

Mean

$$\mu = E(x) = \sum_{x=0}^{\infty} x f(x) = \sum_{x=0}^{\infty} x \cdot \frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=1}^{\infty} \frac{\lambda^x e^{-\lambda}}{(x-1)!}$$

$$= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = \lambda e^{-\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots\right]$$

$$= \lambda e^{-\lambda} \cdot e^{\lambda} = \underline{\lambda} \qquad \because e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

Variance

Mean, $\mu = \lambda$

Variance, $\sigma^2 = \lambda^2$

Q. Fit a poisson distribution to the set of observations.

x	0	1	2	3	4
f	122	60	15	2	1

→ Mean $= \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{(0\times122)+(1\times60)+(2\times15)(3\times2)+(4\times1)}{122+60+15+2+1}$ $\Sigma f_i =$

$= 0.5$

In poisson distribution, mean $= \lambda = 0.5$

Hence, the theoretical frequencies / expected frequencies for 'x' successes is $N \cdot \frac{\lambda^x e^{-\lambda}}{x!} = 200 \cdot \frac{(0.5)^x e^{-0.5}}{x!}$

i.e for x: 0 1 2 3 4

f: 121.3 60.65 15.16 2.52 0.31

rounding off → 121 61 15 2 0

Q. In a given city 6% of all drivers get atleast one parking ticket per year. Determine the probabilities that among 80 such drivers

(a) 4 will get atleast one parking ticket in any given year.

(b) Atleast 3 will get atleast one parking ticket in any given year.

(c) Anywhere from 3 to 6 inclusive get atleast one parking ~~ticking~~ ticket in any given year.

→ Success - getting atleast one parking ticket per given year.

let X - the no. of successes.

$n = 80$ $p = 0.06$ $q = 1-p = 0.94$

note: Since poisson distribution is a limiting case of B.D, $\lambda = np$

$\lambda = np = 80 \times 0.06 = 4.8$

$F(x) = \frac{\lambda^x e^{-\lambda}}{x!}$, $x = 0, 1, 2, \ldots$

$$F(x) = \frac{(4\cdot8)^x \, e^{-4\cdot8}}{x!}$$

(a) $P(X=4) = f(4) = \frac{(4\cdot8)^4 \, e^{-4\cdot8}}{4!} = 0.182$

(b) $P(\text{atleast } 3) = P(X \geq 3) = 1 - P(X<3) = 1 - [P(X=2) + P(X=1) + P(X=0)]$

$$= 1 - \left[\frac{(4\cdot8)^2 e^{-4\cdot8}}{2!} + \frac{4\cdot8^1 \, e^{-4\cdot8}}{1!} + \frac{4\cdot8^0 \, e^{-4\cdot8}}{0!}\right]$$

$$= 0.857$$

(c) $P(3 \text{ to } 6) = P[3 \leq X \leq 6] = P(X=3) + P(X=4) + P(X=5) + P(X=6)$

$$= \frac{4\cdot8^3 \, e^{-4\cdot8}}{3!} + \frac{4\cdot8^4 \, e^{-4\cdot8}}{4!} + \frac{4\cdot8^5 \, e^{-4\cdot8}}{5!} + \frac{4\cdot8^6 \, e^{-4\cdot8}}{6!}$$

$$= 0.648$$

Q. Show that poision distribution is a limiting case of binomial distribution.

→ I (i) No. of trials 'n' is extremely large i.e $n \to \infty$

(ii) The probability of sucess p is very small i.e $p \to 0$

(iii) $\lambda = np$

$p = \lambda/n$, $q = 1-p = 1 - \frac{\lambda}{n}$

II The Binomial probability mass function (pmf), $P(X=x) = {}^nC_x \, p^x \, q^{n-x}$

$$= \frac{n!}{x!\,(n-x)!} p^x (1-p)^{n-x} = \frac{n(n-1)(n-2)\cdots n-(x-1)(n-x)!}{x!\,(n-x)!} p^x (1-p)^{n-x}$$

$$= \frac{n(n-1)(n-2)\cdots n-(x-1)}{x!} p^x \cdot (1-p)^{n-x}$$

III Replacing p with λ/n.

$$P(X=x) = \frac{n(n-1)(n-2)\cdots(n-(x-1))}{x!} \left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n \cdot n\left(1-\frac{1}{n}\right) \cdot n\left(1-\frac{2}{n}\right) \cdots n\left(1-\frac{(x-1)}{n}\right)}{x!} \cdot \frac{\lambda^x}{n^x} \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= n^x \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \cdots \left(1-\frac{(x-1)}{n}\right) \cdot \frac{\lambda^x}{n^x} \left(1-\frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{\lambda^x}{x!} \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \cdots \left(1-\left(\frac{x-1}{n}\right)\right) \cdot \frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^x}$$

IV Taking the limit as $n \to \infty$

$$\lim_{n\to\infty} P(X=x) = \frac{\lambda^x}{x!} \lim_{n\to\infty} \frac{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \cdots \left(1-\left(\frac{x-1}{n}\right)\right) \cdot \left(1-\frac{\lambda}{n}\right)^n}{(1-\lambda/n)^x}$$

$\lim_{n\to\infty} \left(1-\frac{1}{n}\right) = 1$

$\lim_{n\to\infty} \left(1-\frac{2}{n}\right) = 1$

[$\because$ x is any fixed no: any term like $\frac{1}{n}, \frac{2}{n}, \cdots$ approaches to zero].

$\lim_{n\to\infty} \left(1-\left(\frac{x-1}{n}\right)\right) = 1$

Also by limit identity, $\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda}$ $\left(\because \lim_{n\to\infty} \left(1+\frac{x}{n}\right)^n = e^x\right)$

& $\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^x = 1$

$$\lim_{n\to\infty} P(X=x) = \frac{\lambda^x}{x!} \left[\frac{1 \cdot 1 \cdot \cdots \cdot 1 \cdot e^{-\lambda}}{1}\right]$$

$$\lim_{n\to\infty} P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$$ is the pmf of poisson distribution.

Normal Distribution

It is a continuous distribution i.e, it measures continuous data where the values can take any number within a range, it can be derived from binomial distribution under the two conditions :-

1) The no. of trials, 'n' is very large.
2) The probability of success is close to ½ (0.5)

Probability density function (pdf)

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

OR

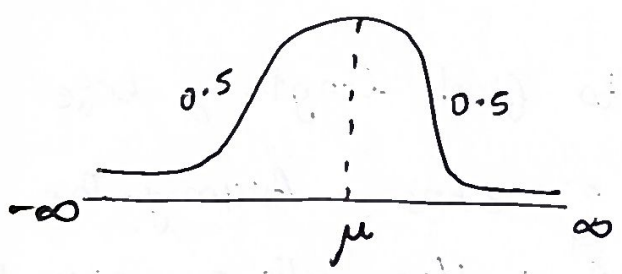
$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) = 1$

If a variable 'x' has normal distr. with mean μ and standard deviation σ, $X \sim N(\mu, \sigma)$.

The probability distribution curve - Normal curve. It is a bell shaped curve.



Symmetrical around its center line. The center divides the area into two equal parts. since the total probability is ~~one~~ 1, ~~And~~ area under RHS is 0.5 and area under L.H.S is 0.5. The mean, median and mode are all equal.

Standard form of Normal Distribution

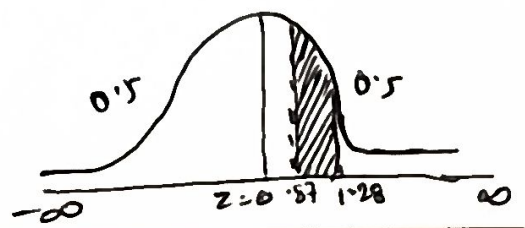
If X is a continuous normal random variable with mean μ & standard deviation σ, it can be transformed into a standardized variable; the standardized normal variable, z is defined as $z = \frac{X-\mu}{\sigma}$ where, the variable z always has mean 0 ($\mu = 0$) and S.D 1 ($\sigma = 0$). Then pdf of z is $f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}z^2}$, $-\infty < z < \infty$

Q. Find the probability that a Random variable having a standard normal distribution will take on the value (a) b/w 0.87 & 1.28

(b) b/w −0.34 & 0.62 (c) greater than 0.85 (d) greater than −0.65

→ (a) $P[0.87 < z < 1.28] =$ ~~$P[0 < z < 1.28]$~~

$= P[0 < z < 1.28] - P[0 < z < 0.87]$

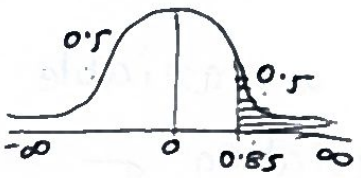


$= 0.3997 - 0.3078 = \underline{0.0919}$

(b) $P[-0.34 < z < 0.62] = P(0 < z < 0.34) + P(0 < z < 0.62)$

$= 0.1331 + 0.2324 = \underline{\underline{0.3655}}$

(c) $P[z > 0.85] = P(z > 0) - P(0 < z < 0.85) = 0.5 - 0.3023 = 0.1977$



(d) $P[z > -0.65] = P(z > 0) + P(0 < z < 0.65)$



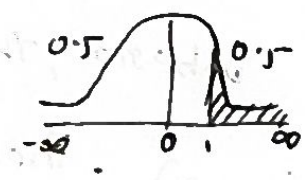
$= 0.5 + 0.2422 = \underline{\underline{0.7422}}$

Q. A sample of 100 dry battery cells tested to find length of life produced the following results $\mu = 12$ hrs, $\sigma = 3$ hrs. Assuming the data to be normally distributed. what % of battery cells are expected to have life (i) more than 15 hrs (ii) less than 6 hrs (iii) b/w 10 and 14 hrs

→ Here X denote the length of life of the dry battery cells.

$$z = \frac{X - \mu}{\sigma} = \frac{X - 12}{3}$$

(i) $P[X > 15]$ ⇒ put $X = 15$, $z = \frac{15-12}{3} = \underline{\underline{1}}$

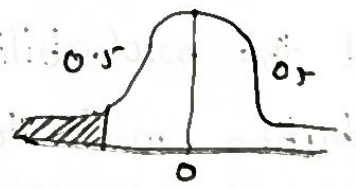


$P(X > 15) = P(z > 1) = P(z > 0) - P(0 < z < 1)$

$= 0.5 - 0.3413 = \underline{\underline{0.1587}}$

(ii) $P(X < 6)$

$X = 6$, $z = \frac{6-12}{3} = -2$



$P(X < 6) = P(X < -2) = P(z < 0) - P(0 < z < 2) = 0.5 - 0.4772 = \underline{\underline{0.0228}}$

(iii)

49.74%

note: when $N<30$, $\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim t_{(n-1)}$ sample mean → $\bar{x}$

when $N \geq 30$, $\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim N(0,1)$

Q. A ~~popola~~ population is known to follow a normal distribution with mean 2 and S.D 3. Find the probability that the mean of sample of size 16 taken from the population will be greater than 2.5.

→ Population – Normal, $n=16$ ~~notei samplevs~~

(μ) mean = 2

(σ) S.D = 3

$P(\bar{x} > 2.5) = ?$

~~$\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$~~ $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$P(\bar{x} > 2.5) = P\left(\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} > \frac{2.5-\mu}{\sigma/\sqrt{n}}\right) = P\left(z > \frac{2.5-2}{3/\sqrt{16}}\right)$

$= P\left(z > \frac{0.5 \times 4}{3}\right) = P(z > 0.67) = 0.2486$ ~~VR~~

$= 0.5 - P(0<z<0.67) = 0.5 - 0.2486 = 0.2514$

Q. A sample of size 25 is taken from a normal formation with $\mu=1$, $\sigma=1.5$. Find the probability that the sample mean is negative.

→ $n=25$, $\mu=1$, $\sigma=1.5$ $P(\bar{x}<0) = ?$

$z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ → $P(\bar{x}<0) = P\left(z > \frac{0-\mu}{\sigma/\sqrt{n}}\right) = P\left(z > \frac{0-1}{1.5/\sqrt{25}}\right)$

$= P\left(z > \frac{-1 \times 5}{1.5}\right) = ~~3.33~~ \; P(z > -3.33) = 0.5 - P(0<z<3.33)$

$= 0.5 - 0.4996 = 0.0004$

Q. A random sample of size 36 is taken from normal population. Find the probability that the sample mean exceeds population mean by atleast 1. SD = 3.

→ $n = 36,\ \mu = 1,\ S.D\ (\sigma) = 3$

(c) $P(\bar{x} > \mu + 1) = P\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}\right) > \frac{\mu + 1 - \mu}{\sigma/\sqrt{n}}$

(d) $= P\left(z > \frac{6}{3}\right) = P(z > 2) = 0.5 - P(0 < z < 2)$

$= 0.5 - 0.4772 = 0.0228$

Q. A random sample of size 12 is taken from a normal population. $N(\mu, 3)$. Find the probability that the variance of the sample lies b/w 3.4 and 14.8.

variance - s^2

→ $n = 12$ $\quad df = 11$ $\quad \alpha = 0.04$

$P(3.4 < s^2 < 14.8)$ multiplying with $\frac{(n-1)}{\sigma^2}$

$\Rightarrow P\left(\frac{n-1}{\sigma^2} \times 3.4 < \frac{(n-1)s^2}{\sigma^2} < \frac{(n-1)}{\sigma^2} \times 14.8\right)$

$\Rightarrow P\left(\frac{12-1}{3^2} \times 3.4 < X^2 < \frac{(12-1)}{3^2} \times 14.8\right) = 4.15 < X^2 < 18.05$

$= P(X^2 > 4.16) - P(X^2 > 18.05) = 0.9650 - 0.0 =$

Q. A random sample of size $N(20, 8)$. Find the prob that the sample variance is less than the population variance.

→ $n = 18,\ \mu = 20,\ \sigma = 8$ $\quad df = 17$

$P(s^2 < \sigma^2) = P\left(\frac{(n-1)s^2}{\sigma^2} < \frac{(n-1)\sigma^2}{\sigma^2}\right) = P(x^2 < n-1) = P(x^2 < 17)$

(i) $\Rightarrow P(x^2 < 17) = 1 - P(x^2 > 17) = 1 - 0.5 = 0.5$

HW

1 Q. A random sample of size 36 is taken from normal population with $\sigma^2 = 1$. Find the probability that the sample mean will differ population mean by atleast 1.

1 Q. A random sample of size 60 is taken from a normal population w/ variance 9. Find the probability than sample mean will be within 0.5 of population mean.

Ans: 1) $n = 36, \quad \sigma = 1$

$P(\bar{x} > \mu \pm 1)$

Test of Hypothesis

1) Construct null hypothesis, H_0 $(\theta = \theta_0)$
2) Construct alternative hypothesis, H_1 $(\theta = \theta_1)$
3) Choose significance level α (0.05 or 0.01)
4) Select a test statistic & calculate value of test statistic
5) Find the table value
6) Reject or accept H_0 based on the calculated & ~~tabulated~~ tabled value

Test for large sample.

Binomial, $\mu = np$, S.D, $\sigma = \sqrt{npq}$

$n > 30$, Binomial $\longrightarrow$ Normal

$X \sim$ Binomial

$z = \frac{X - \mu}{\sigma} \sim$ Normal

1) $|z| < 1.96$, accept H_0 with 5% LOS (Level of Significance)
2) $|z| > 1.96$, reject the H_0 with 5% level
3) $|z| > 2.58$, reject H_0 with 1%

Q. A coin was tossed 400 times and the head turned up 216 times. Test ~~Thus~~ the hypothesis that the coin is unbiased at 5% level of significance.

→ $H_0 \Rightarrow$ coin is unbiased

P (getting a tail) = $\frac{1}{2}$ = P (success)

Expected no. of successes = $\frac{1}{2} \times 400 = 200$

Observed no. of success = 216

The excess of no. of success from observed and expected = 216 - 200 = 16

S.D, $\sigma = \sqrt{npq} = \sqrt{400 \times \frac{1}{2} \times \frac{1}{2}} = 10$

$Z = \frac{X - 1}{\sqrt{npq}} = \frac{X - np}{\sqrt{npq}}$ → diff b/w obs & exp $= \frac{16}{10} = 1.6$

$|z| = 1.6 < 1.96 \Rightarrow$ accept H_0 i.e coin is unbiased.

Q. A die was thrown 9000 ~~th~~ times and a throw of 5 or 6 was obtained ~~9000~~ 3240 times. On the assumption of random throwing that data indicate an unbiased die

→ H_0: die is unbiased

P (5 or 6) = P (success) = $\frac{2}{6} = \frac{1}{3}$

Expected no. of success = $\frac{1}{3} \times 9000 = 3000$

Observed no. of success = 3240

Excess of no. of success from the observed and expected $(x - np)$ =

$= 3240 - 3000 = \underline{240}$

$\mu, \sigma = \sqrt{npq} = 44.72$

$z = \frac{x - \mu}{\sigma} = \frac{240}{44.72} = 5.37$

$|z| = 5.37 > 2.58 \Rightarrow$ Test is significant means die is biased.

Testing of single mean

step 1: $H_0 : \mu = k$

step 2: $H_1 : \mu \neq k$ or $\mu < k$ or $\mu > k$

step 3: choose level of significance (0.05 or 0.01)

step 4: Test statistics (i) $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$, σ known

step 5: Z_α $\quad H_1 : \mu \neq k$ (two tailed test)

Step 6: conclusion $\Rightarrow$ when $|z$ calculated$| > Z_\alpha$, we reject H_0

when $z_{calculated} < -Z_\alpha$, reject H_0 (left tailed test)

when $z_{calculated} > Z_\alpha$, reject H_0 (right tailed test)

Q. The mean wt obtained from a random sample of 100 is 64g. The S.D of wt distribution of the population is 3g. Test the statement that mean wt of population is 67g. at 5% LOS.

→ Given, $\bar{x} = 64$, $n = 100$, $\sigma = 3$

1) H_0: mean weight is 67 g i.e $H_0 : \mu = 67$

2) $H_1 : \mu \neq 67$

3) $\alpha = 0.05$

4) $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = -10$ ($z_{calculated}$)

5) $Z_\alpha = 1.96$

6) Conclusion, $Z_{calculated} = -10 < Z_\alpha = 1.96$

here $|z| = 10 > Z_\alpha = 1.96$ ⇒ we reject H_0. as $\mu = 67$

Q. The avg mark in maths of a sample of 100 students was 51. with a S.D of 6 marks. Could this have been a random sample from population w/ avg marks 50. Test at 1% LOS.

→ $n = 100$, $\bar{x} = 51$, $\sigma = 6$

1) $H_0 : \mu = 50$

2) $H_1 : \bar{x} \neq 50$

3) $\alpha = 0.01$

$\alpha = 0.01$

⇒ $Z_\alpha = 2.58$ (two tailed)

4) $z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{51 - 50}{6/\sqrt{100}} = 1.67$

5) $Z_{(\alpha = 0.01)} = 2.58$

6) $Z_{calculated} = 1.67 < Z_\alpha = 2.58$ accept H_0 as $\mu = 50$

∴ Sample is taken population with mean 50.

Hw

Q. A normal population has mean 6.8 , S.D = 1.5 . A sample of 400 members have mean of 6.75 . Is the difference significant.

Q. The mean life of 100 fluorescent lights tube produced by a company is computed to be 1570 hours with S.D 120 hours. The company claims that avg life of tubes produced by company is 1600 hours using LOS 0.05. Is the claim acceptable?

Test concerning two means

1) $H_0 : \mu_1 = \mu_2$

2) $H_1 : \mu_1 \neq \mu_2$ or $\mu_1 < \mu_2$ or $\mu_1 > \mu_2$

3) choose LOS

4) Test statistic (i) $Z = \dfrac{\bar{x_1} - \bar{x_2}}{\sqrt{\dfrac{\sigma_1^2}{2n_1^2} + \dfrac{\sigma_2^2}{2n_2^2}}}$ when σ_1 & σ_2 are known

(ii) $z = \dfrac{\bar{x_1} - \bar{x_2}}{\sqrt{\dfrac{s_1^2}{2n_1} + \dfrac{s_2^2}{2n_2}}}$, when σ_1, σ_2 unknown

5) Z_α (table value)

6) Reject or accept H_0

Q. The avg income of persons was ₹ 210 w/ SD ₹ 10 in sample of 100 people of a city. For another sample of 150 persons, the avg income was ₹ 220 w/ S.D of 12. The SD of incomes of the people of the city was ₹ 11. Test whether there is any significant difference b/w the avg incomes of the localities.

→ $n_1 = 100$, $n_2 = 150$, $\bar{x_1} = 210$, $\bar{x_2} = 220$, $S_2 = 10$

$S_2 = 12$

$H_0 : \bar{x_1} = \bar{x_2}$

$H_1 : \bar{x_1} \neq \bar{x_2}$

$\alpha : 0.05$

$$z = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} = -7.14$$

$Z_\alpha = 1.96$

$Z_{calculated} = -7.14 < Z_\alpha = 1.96$

for two tailed, $|Z_{calculated}| = 7.14 > Z_\alpha = 1.96 \Rightarrow$ Reject H_0

$\therefore$

Q. From the data given below compute standard error of the diff of two sample means & find out if the two means significantly differ at 5% LOS.

	no. of Items	mean	S.D
Group 1	50	181.5	3.0
Group 2	75	179	3.6

note: For two tailed test of the form $H_1 : \bar{x_1} \neq \bar{x_2}$

(i) case 1 : $\alpha = 0.01$, $Z_\alpha = 2.58$

(ii) case 2 : $\alpha = 0.05$, $Z_\alpha = 1.96$

For one tailed test of form : $H_1 : \bar{x_1} < \bar{x_2}$ or $\bar{x_1} > \bar{x_2}$

(i) for $\alpha = 0.01$, $Z_\alpha = 2.33$

(ii) $\alpha = 0.05$, $Z_\alpha = 1.645$

Q. Individual test were given to two groups of boys and girls.

	Mean	SD	size
Girls	75	8	60
Boys	73	10	100

Examine difference b/w mean scores is significant.

$\rightarrow H_0 : \bar{x_1} = \bar{x_2}$

$H_1 : \bar{x_1} \neq \bar{x_2}$

$$Z = \frac{\bar{x_1} - \bar{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = 1.39$$

$|z_{calculated}| = 1.39$

$\alpha = 0.05 \Rightarrow z_\alpha = 1.96$

$|z| = 1.39 < 1.96 = z_\alpha \Rightarrow$ we accept $H_0 \Rightarrow \bar{x}_1 = \bar{x}_2$

no significant difference b/w the means.

Test concerning two population variation

$H_0 : S_1 = S_2$

1) $H_0 : \sigma_1 = \sigma_2$

2) $H_1 = \sigma_1 \neq \sigma_2$

3) choose α (0.05 or 0.01)

4) (i) $$z = \frac{S_1 - S_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}}$$ σ_1, σ_2 known

(ii) $$z = \frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_1} + \frac{S_2^2}{2n_2}}}$$ σ_1, σ_2 unknown

5) z_α

6) conclusion (accept or reject)

Q. Random samples of population following data

	country A	country B
mean height (inches)	67.42	67.25
SD	2.58	2.50
no. of sample	1000	1200

Is the difference b/w S.D significant.

→ 1) $H_0 : \sigma_1 = \sigma_2$

2) $H_1 = \sigma_1 \neq \sigma_2$

3) $\alpha = 0.05$

4) $z = \dfrac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = 1.03$

$s_1 = 2.58$

$s_2 = 2.58$

$n_1 = 1000$

$n_2 = 1200$

5) $z_\alpha = 1.96$

6) $|z| = 1.03 < 1.96 = z_\alpha$ we accept $H_0 \Rightarrow \sigma_1 = \sigma_2$

$\therefore$ no significant difference b/w σ_1 & σ_2

Test concerning <u>single</u> population variance.

1) $H_0: \sigma = \sigma_0$

2) $H_1 = \sigma \neq \sigma_0$ or $\sigma < \sigma_0$ or $\sigma > \sigma_0$

3) α

4) $z = \dfrac{s - \sigma}{\sigma^2/\sqrt{2n}}$

5) z_α

6) conclusions $|z_{calc}| > z_\alpha \Rightarrow$ we reject null H_0,

otherwise we accept it

Q. The SD of a sample of size ~~25~~ 50 is 3.6 examine whether the sample was taken from population w/ S.D = 3.3. ($\alpha = 1\%$)

→ $n = 50$, $s = 3.6$

1) $H_0: \sigma = 3.3$

2) $H_1: \sigma \neq 3.3$

3) $\alpha = 0.01$

4) $z = \dfrac{s - \sigma}{\sigma/\sqrt{2n}} = 0.9$

5) $z_\alpha = 2.58$

6) $|z| = 0.9 < z_\alpha = 2.58 \Rightarrow$ accept the null hypothesis H_0,

$\therefore \sigma = 3.3$ $\therefore$ the sample is taken from a population with SD = 3.3.

Small sample test ($n < 30$)

1) χ^2 (chi) test

2) $H_0 : \sigma^2 = \sigma_0 \quad n < 30$

2) $H_1 : \sigma^2 > \sigma_0$

3) α

4) $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$

5) $\chi^2_{(n-1)}{}^{(\alpha)}$, chi square table value

6) conclusion whether to accept or reject H_0

$\chi^2_{calculated} > \chi^2_{(n-1)}{}^{(\alpha)}$, we reject H_0 otherwise, we accept it.

Q. A Random sample of size 18 is taken from a Normal distribution

Test the hypothesis that $H_0 : \sigma^2 = 0.36$ against

$H_1 : \sigma^2 > 0.36$

$s^2 = 0.68, \quad \alpha = 0.05$

→ $n = 18, \quad \chi^2 = \frac{(n-1)s^2}{\sigma^2} = 32.1$

$\chi^2_{(n-1)}{}^{(\alpha)} = \chi^2_{(17)}{}^{(0.05)} = 27.587$ ← from log table

$\chi^2_{calc} > \chi^2_{(n-1)}{}^{(\alpha)} \Rightarrow$ reject H_0.